Updated January 28, 2026

Homework problems for AMAT 300 (Intro to Proofs), Spring 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 2/4): Let $A = \{20a - 26b \mid a, b \in \mathbb{Z}\}$ and $B = \{2c \mid c \in \mathbb{Z}\}$. Prove that $A = B$. (Do it directly, don't use any number theory tricks you might happen to know.)

Problem 2 (due Weds 2/4): Prove that the cardinality of $\{1, \{1\}, \{1, \{1\}\}, \{1, 1\}, \{\{1\}, 1\}\}$ is three. (You need to argue that it has exactly three elements, no more, no less.)

Problem 3 (due Weds 2/4): Let A and B be non-empty finite sets such that $|\mathcal{P}(A \times B)| = |\mathcal{P}(A) \times \mathcal{P}(B)|$. Prove that A and B each have exactly two elements. (Careful not to *assume* they each have exactly two elements, you have to *prove* that they do.)

Problem 4 (due Weds 2/11):

Problem 5 (due Weds 2/11):

Problem 6 (due Weds 2/11):
